

ΨΗΦΙΑΚΗ ΕΠΕΞΕΡΓΑΣΙΑ ΣΗΜΑΤΩΝ

Σχολη ΗΜ&ΜΥ, ΕΜΠ

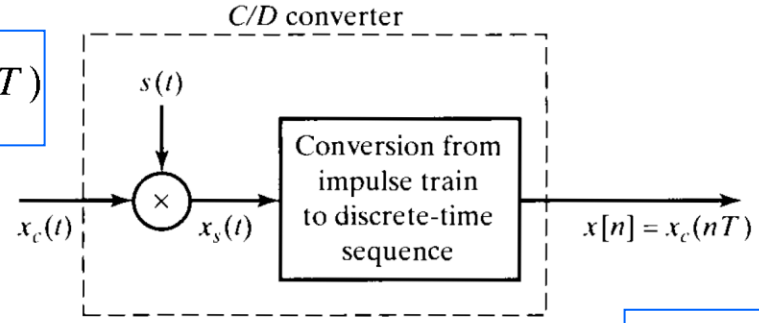
Καθηγητής Πέτρος Μαραγκός

Δειγματοληψία και Ψηφιακή Επεξεργασία Αναλογικών Σημάτων

http://cvsp.cs.ntua.gr/courses/dsp

Μετατροπή Σήματος Συνεχούς Χρόνου σε Διακριτό

$$s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$



(a)

$$x_s(t) = \sum_{n=-\infty}^{\infty} x_c(nT)\delta(t - nT)$$

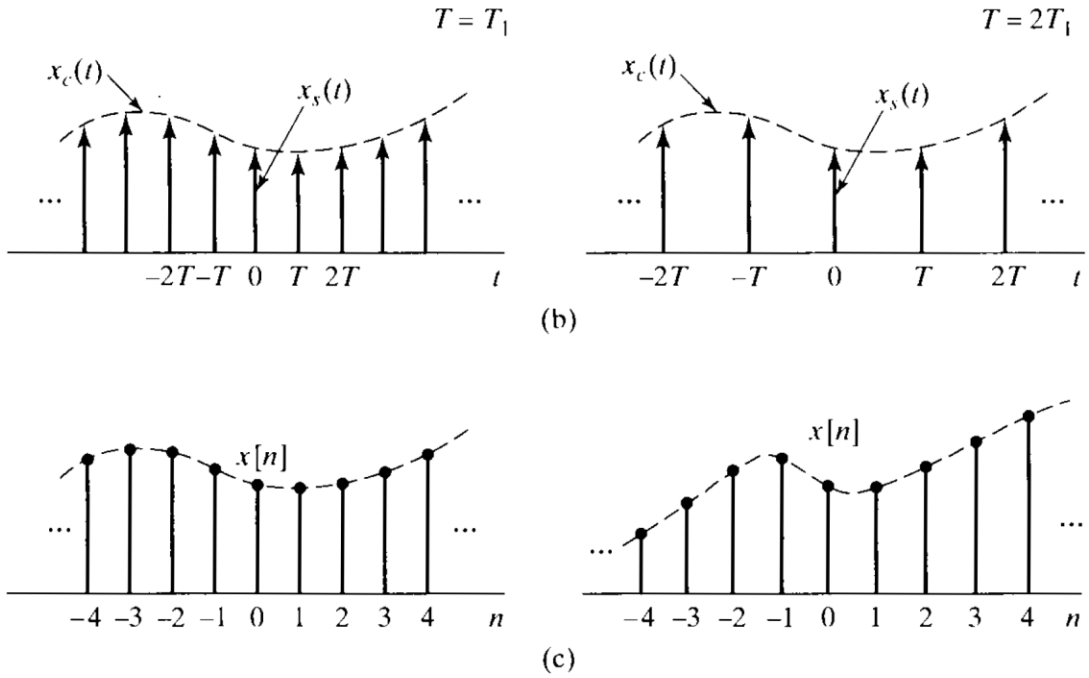
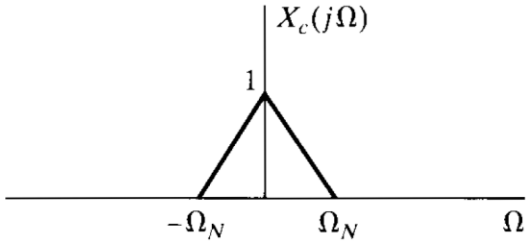
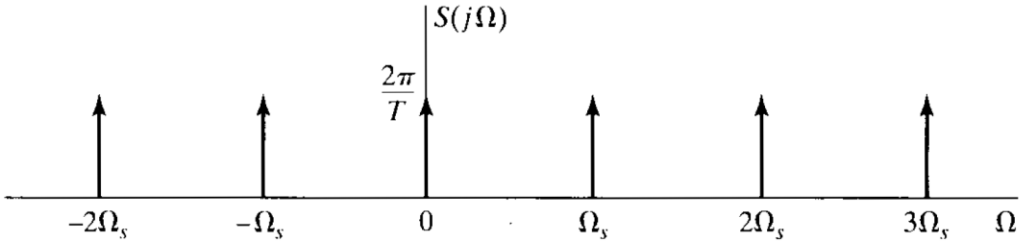


Figure 4.2 Sampling with a periodic impulse train followed by conversion to a discrete-time sequence. (a) Overall system. (b) $x_s(t)$ for two sampling rates. (c) The output sequence for the two different sampling rates.

Ανάλυση Δειγματοληψίας στο Πεδίο της Συχνότητας

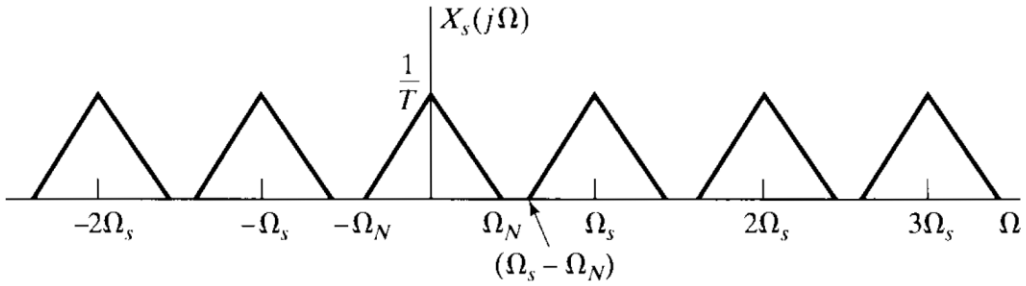


(a)



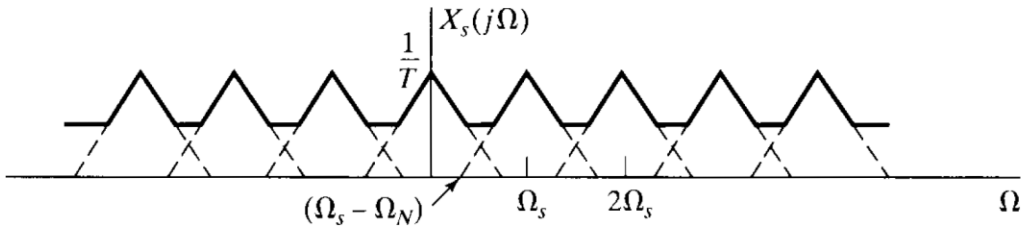
(b)

$$S(j\Omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\Omega - k\Omega_s)$$



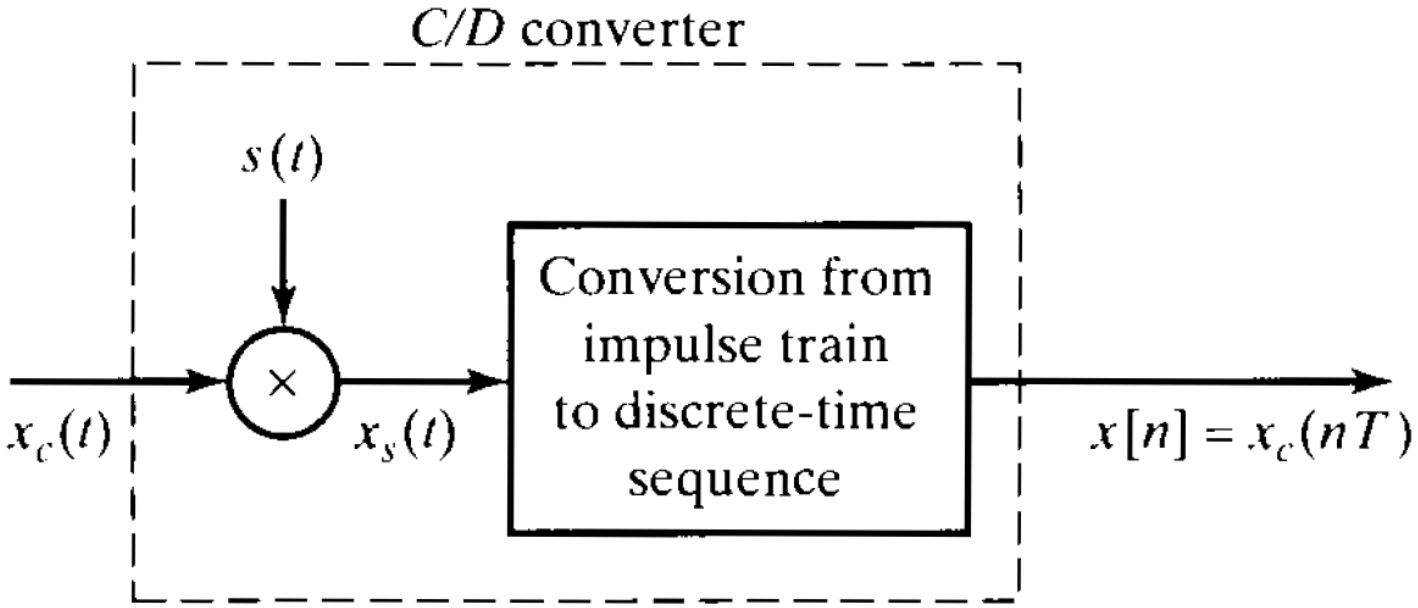
(c)

$$X_s(j\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s))$$



(d)

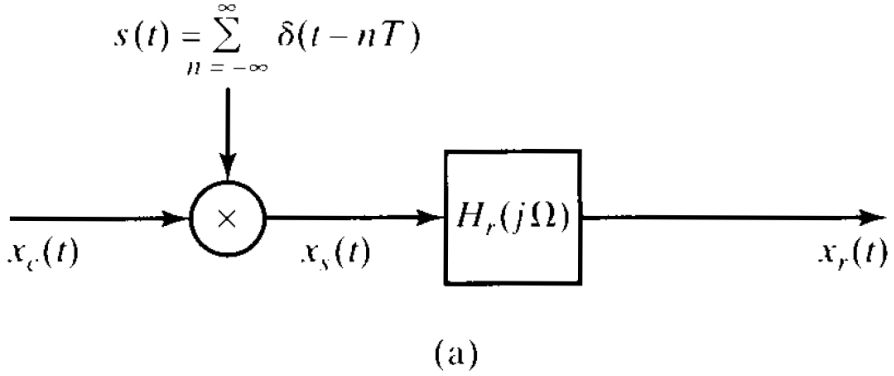
Σ/Δ Μετατροπή Σήματος στα Πεδία Χρόνου & Συχνότητας



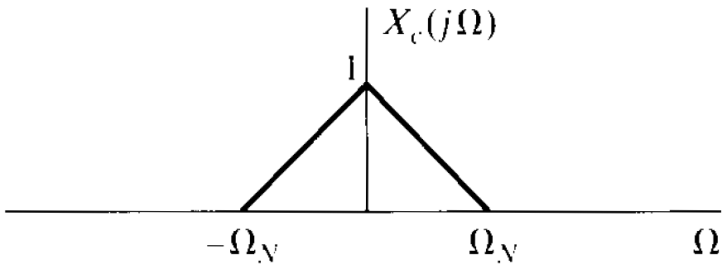
$$s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT) \quad \overset{\mathcal{F}}{\leftrightarrow} \quad S(j\Omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\Omega - k\Omega_s)$$

$$x_s(t) = \sum_{n=-\infty}^{\infty} x_c(nT)\delta(t - nT) \quad \overset{\mathcal{F}}{\leftrightarrow} \quad X_s(j\Omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s))$$

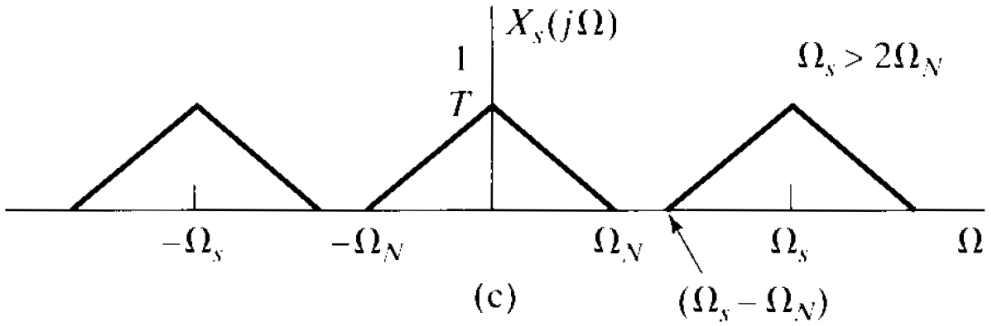
Δειγματοληψία και Ανακατασκευή (Πεδίο Συχνότητας)



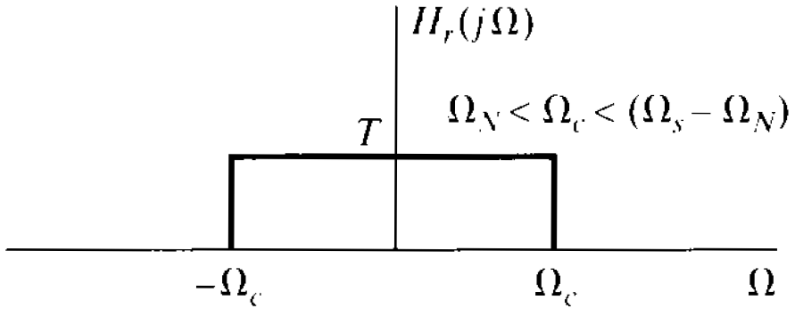
(a)



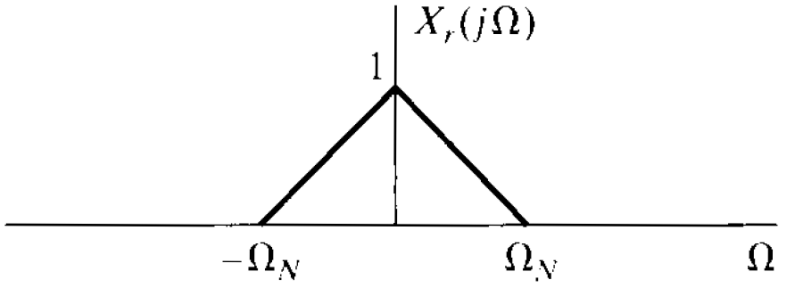
(b)



(c)

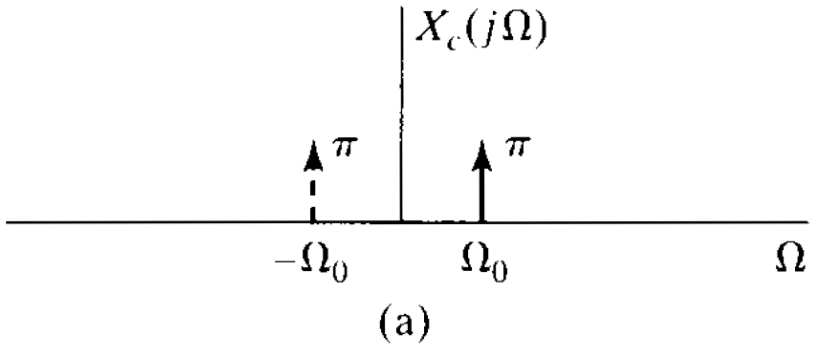


(d)

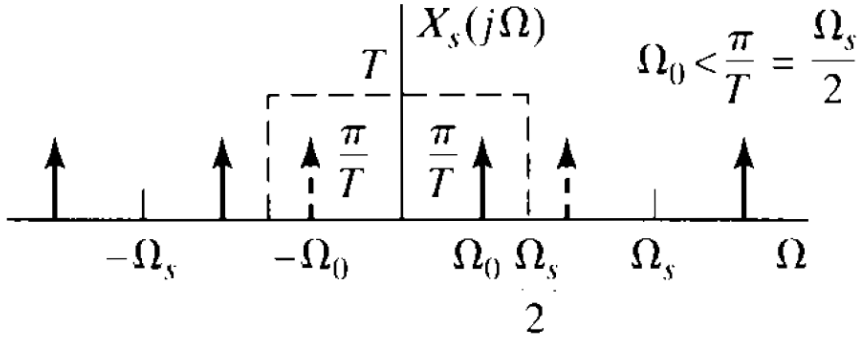


(e)

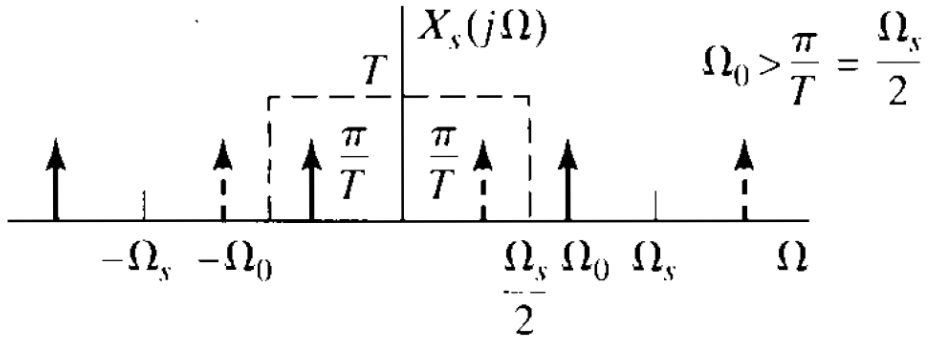
Δειγματοληψία Ημιτονοειδούς (Πεδίο Συχνότητας)



(a)

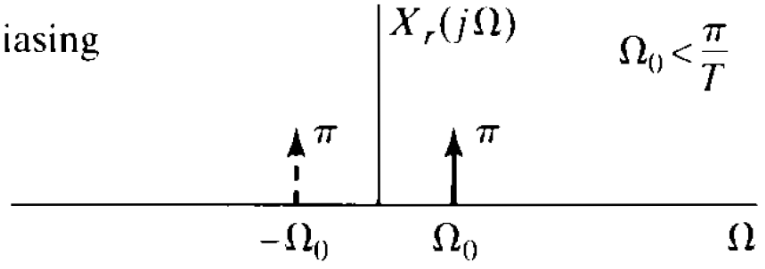


(b)



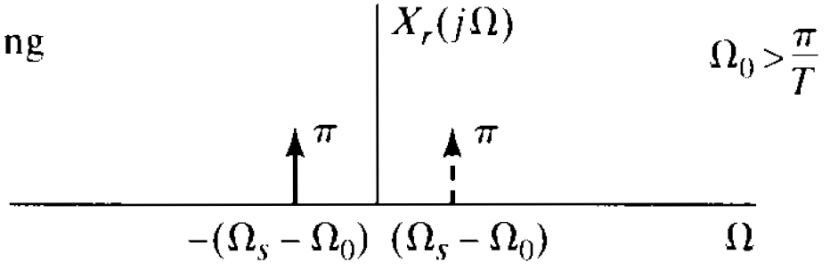
(c)

No aliasing



(d)

Aliasing



(e)

Σχέση Φασμάτων Αναλογικού, Δειγματοληπτημένου και Διακριτού Σήματος

$$x_s(t) = \sum_{n=-\infty}^{\infty} x_c(nT)\delta(t - nT)$$

Φάσμα Δειγματοληπτημένου σήματος

$$X_s(j\Omega) = \sum_{n=-\infty}^{\infty} x_c(nT)e^{-j\Omega Tn}$$

$$x[n] = x_c(nT)$$

Φάσμα Διακριτού σήματος

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

Ερμηνεία 1: DTFT

Από τις παραπάνω σχέσεις προκύπτει ότι:

$$X_s(j\Omega) = X(e^{j\omega})|_{\omega=\Omega T} = X(e^{j\Omega T})$$

Συνεπώς θα είναι:

$$X(e^{j\Omega T}) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s))$$

$$X(e^{j\omega}) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega}{T} - \frac{2\pi k}{T}\right)\right)$$

Ερμηνεια 2: Περιοδική Επανάληψη Αναλογικού Φάσματος

Παράδειγμα Ι: Δειγματοληψία Ημιτονοειδούς

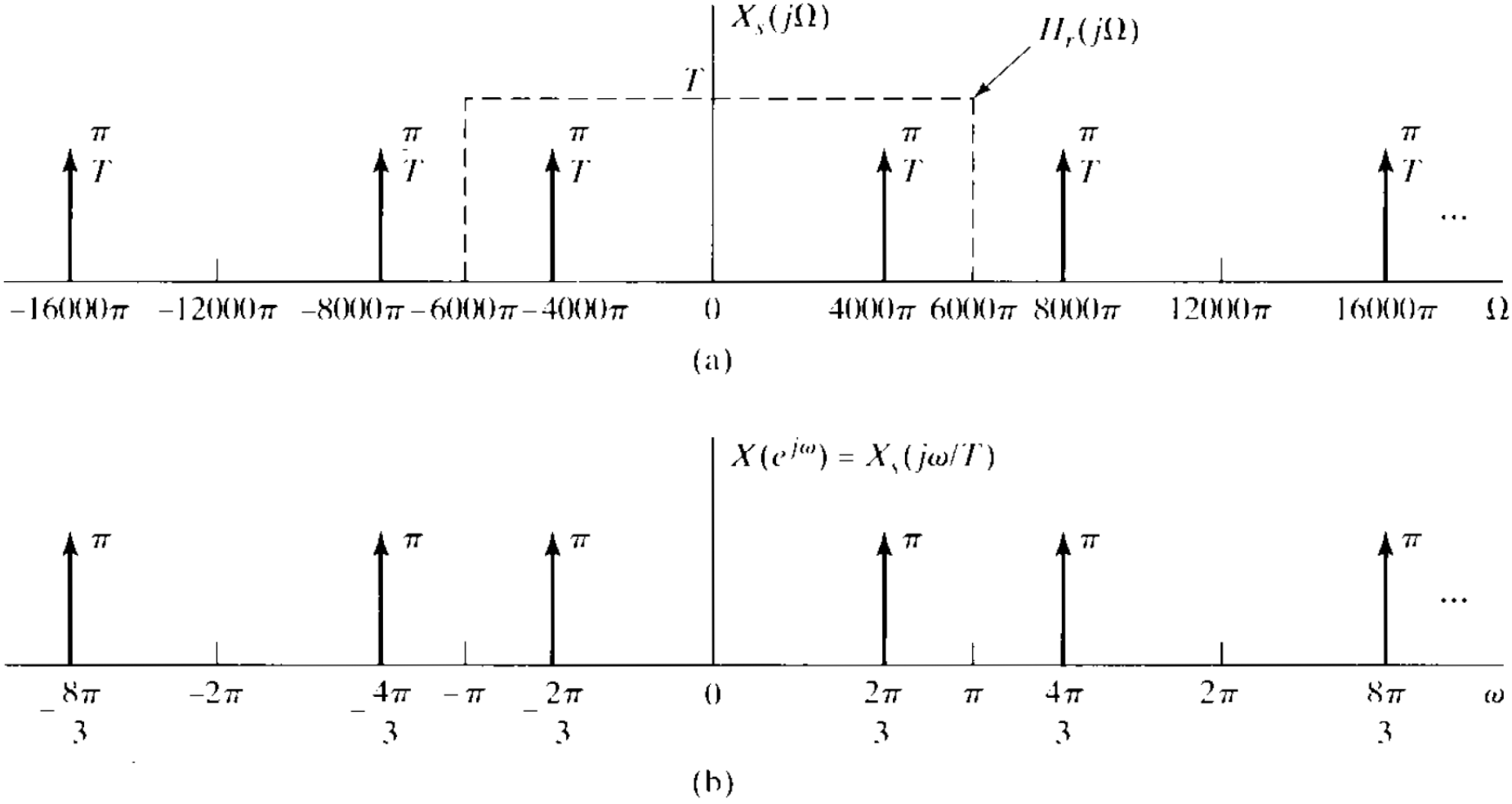


Figure 4.6 Continuous-time (a) and discrete-time (b) Fourier transforms for sampled cosine signal with frequency $\Omega_0 = 4000\pi$ and sampling period $T = 1/6000$.

Παράδειγμα ΙΙ: Δειγματοληψία Ημιτονοειδούς

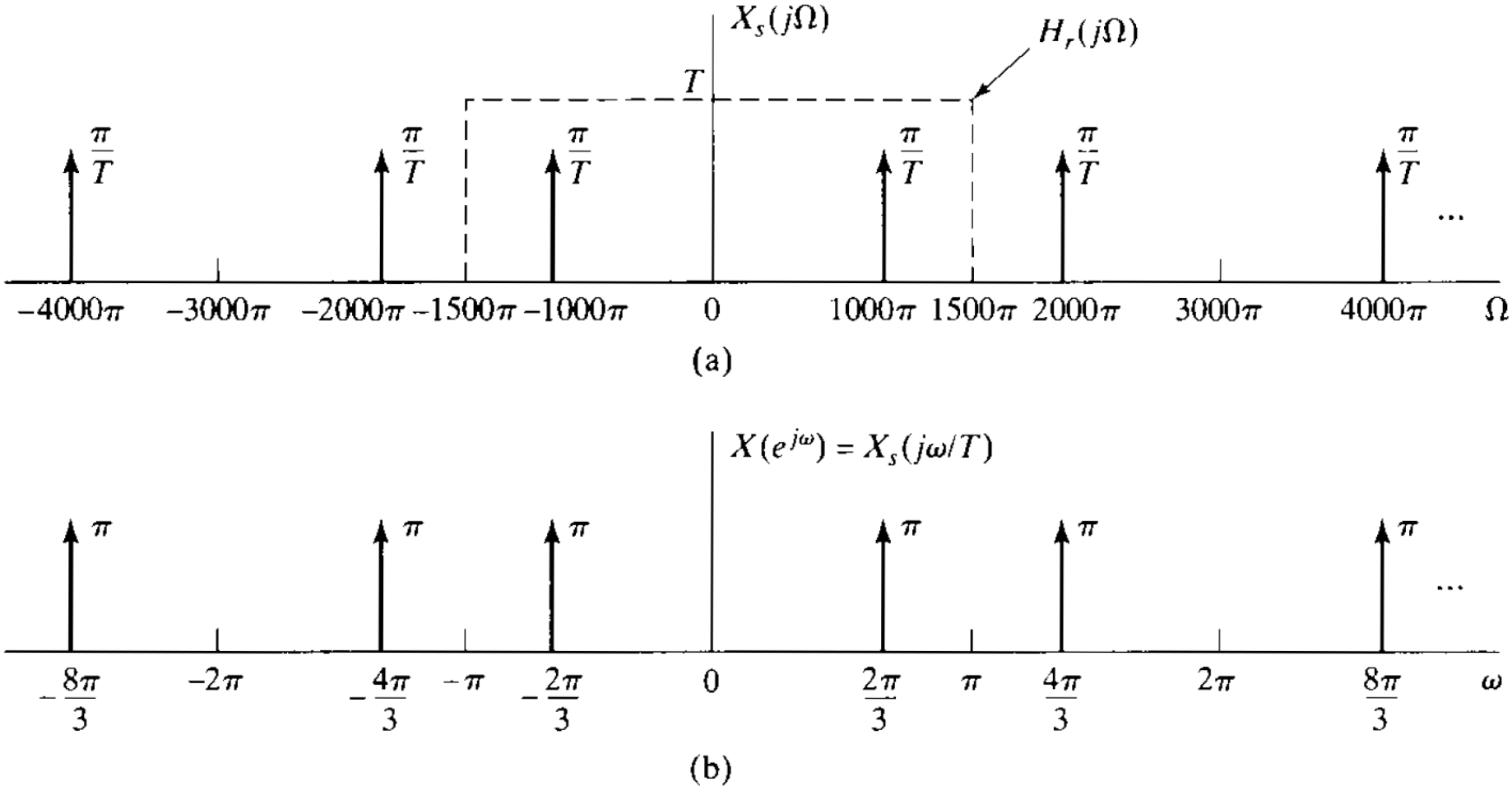
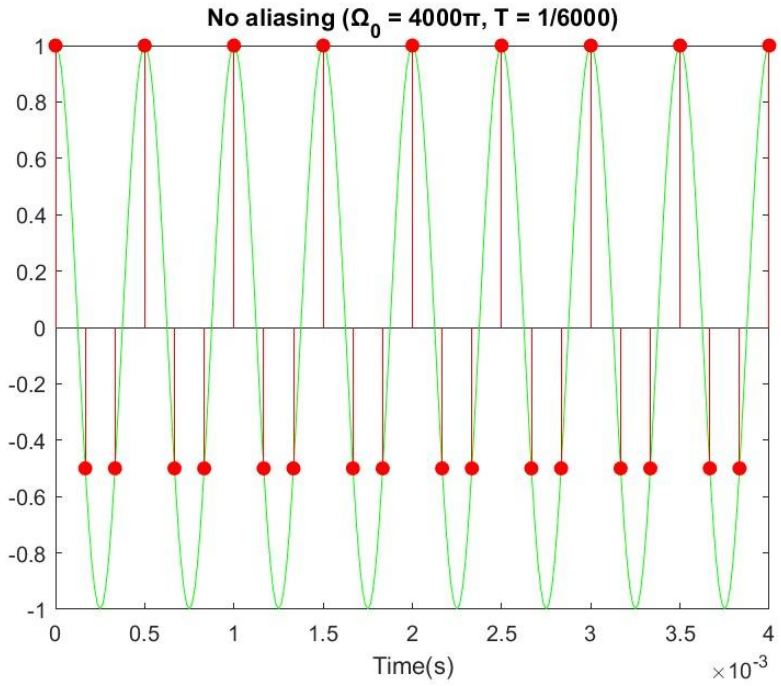
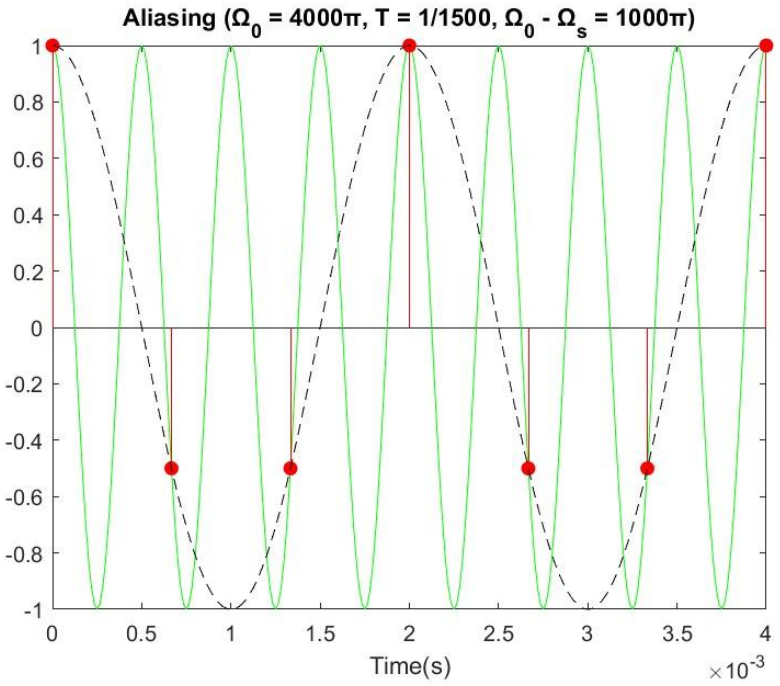


Figure 4.7 Continuous-time (a) and discrete-time (b) Fourier transforms for sampled cosine signal with frequency $\Omega_0 = 4000\pi$ and sampling period $T = 1/1500$.

Δειγματοληψία Ημιτονοειδούς





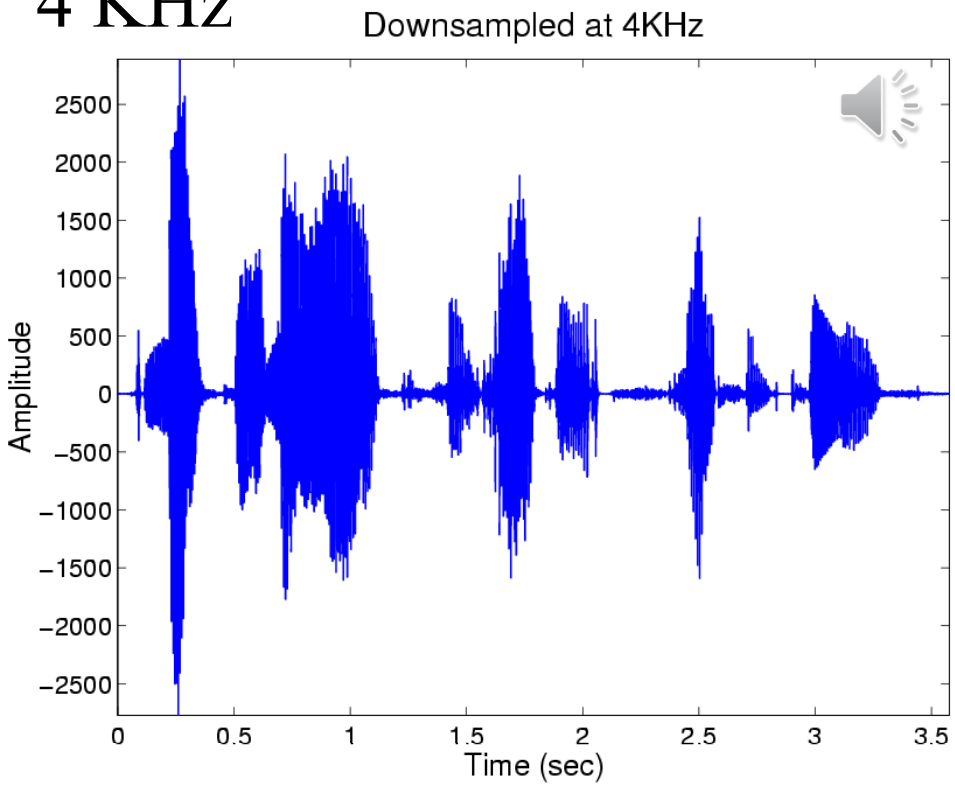


Οι δύο τόνοι στα 2000Hz και 500Hz

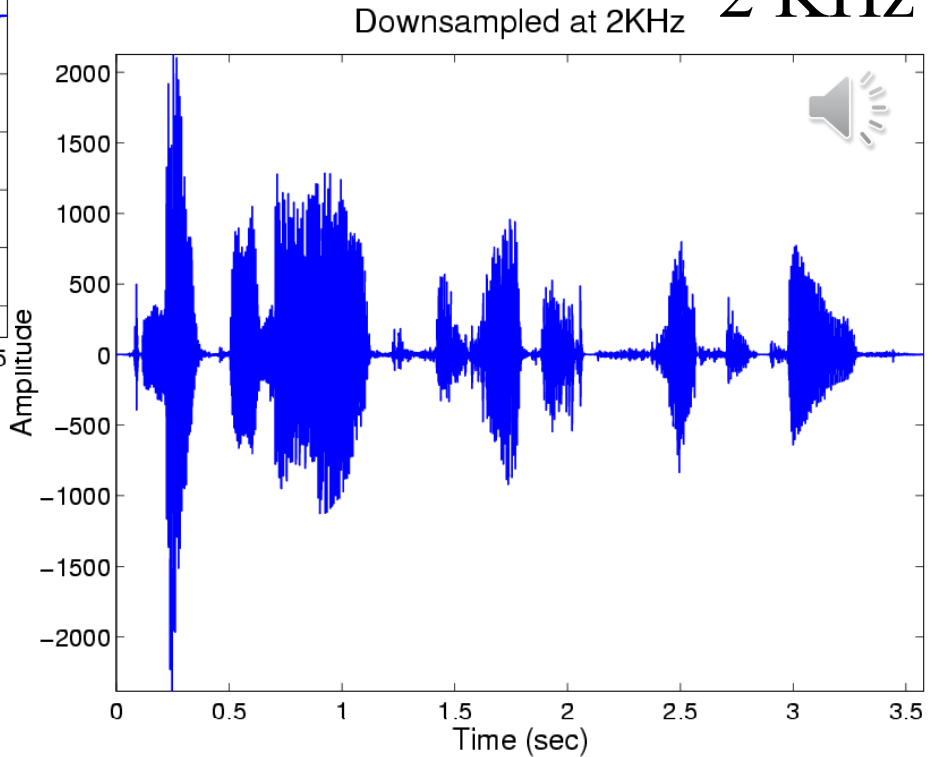
Aliasing σε σήματα φωνής

Original at 16 KHz

4 KHz



2 KHz



Aliasing σε εικόνες



300 x 300



150 x 150



75 x75



38 x38









Θεώρημα Δειγματοληψίας Nyquist-Shannon

Έστω $x_c(t)$ ένα σήμα πεπερασμένου εύρους ζώνης, δηλαδή,

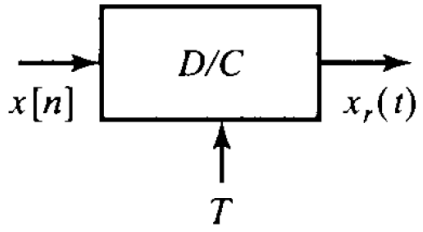
$$X_c(j\Omega) = 0, \text{για } |\Omega| \geq \Omega_N.$$

Τότε, το $x_c(t)$ μπορεί να ανακατασκευαστεί πλήρως από τα δείγματά του, $x_c(nT) = x[n], n \in \mathbb{Z}$, αν για τη συχνότητα δειγματοληψίας, $\Omega_s = 2\pi/T$, ισχύει:

$$\Omega_s \geq 2\Omega_N.$$

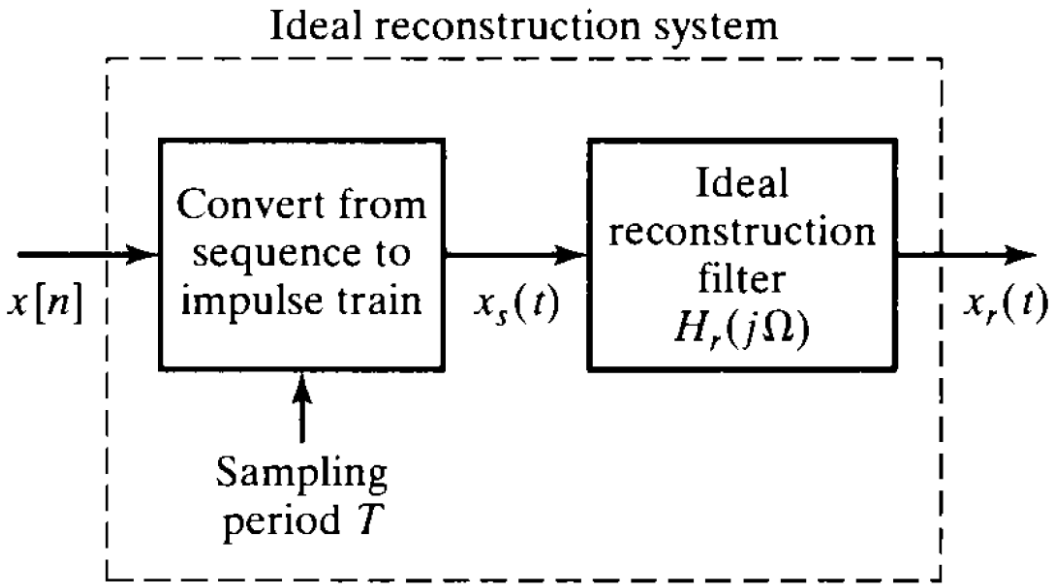
$\boldsymbol{\Omega_N}$: Nyquist frequency
$\boldsymbol{2\Omega_N}$: Nyquist rate

Ιδανική Δειγματοληψία και Ανακατασκευή



$$x[n] = x_c(nT)$$

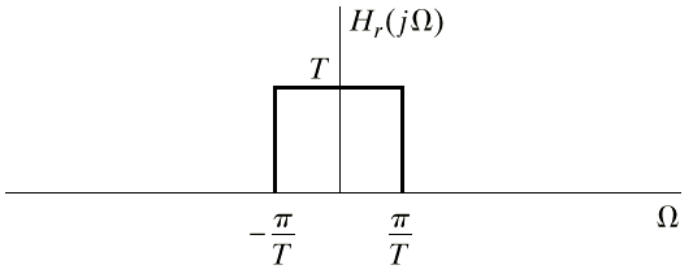
$$x_s(t) = \sum_{n=-\infty}^{\infty} x_c(nT)\delta(t - nT)$$



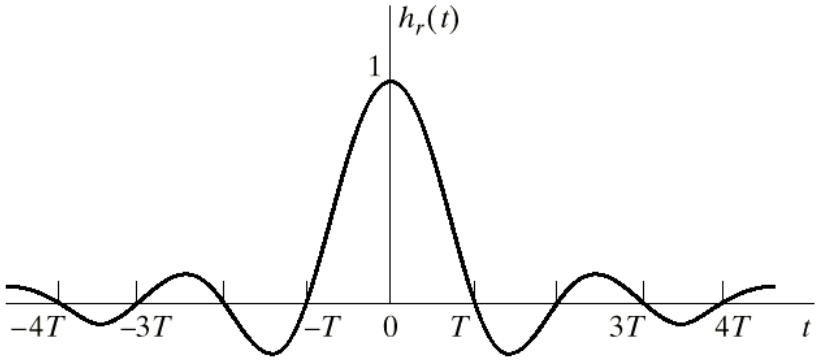
$$x_r(t) = x_s(t) * h_r(t)$$

$$x_r(t) = \sum_{n=-\infty}^{\infty} x[n]h_r(t - nT)$$

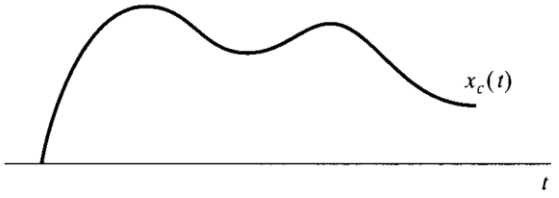
Σ/Δ: Ιδανικό Φίλτρο Ανακατασκευής



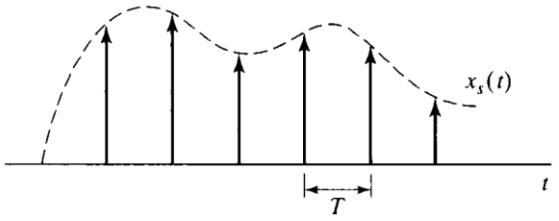
(b)



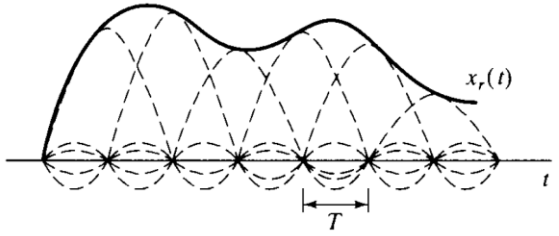
(c)



(a)



(b)



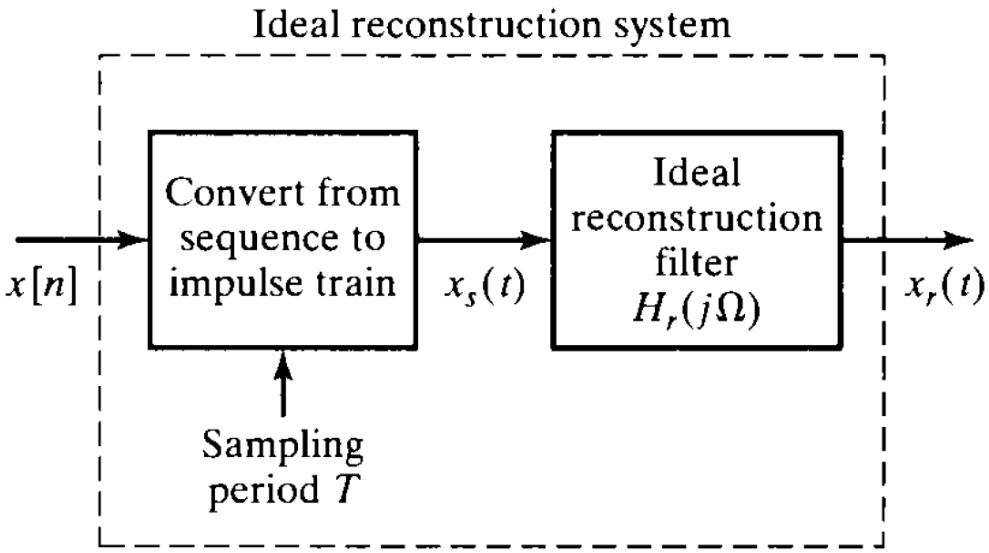
(c)

$$h_r(t) = \frac{\sin(\pi t / T)}{\pi t / T} = \mathrm{sinc}(t/T)$$

$$x_r(t) = \sum_{n=-\infty}^{+\infty} \underbrace{x_c(nT)}_{x[n]} \frac{\sin(\pi(t-nT)/T)}{\pi(t-nT)/T}$$

If $x_c(t)$ is bandlimited and $\Omega_s > 2\Omega_N$, then $x_r(t) = x_c(t)$.

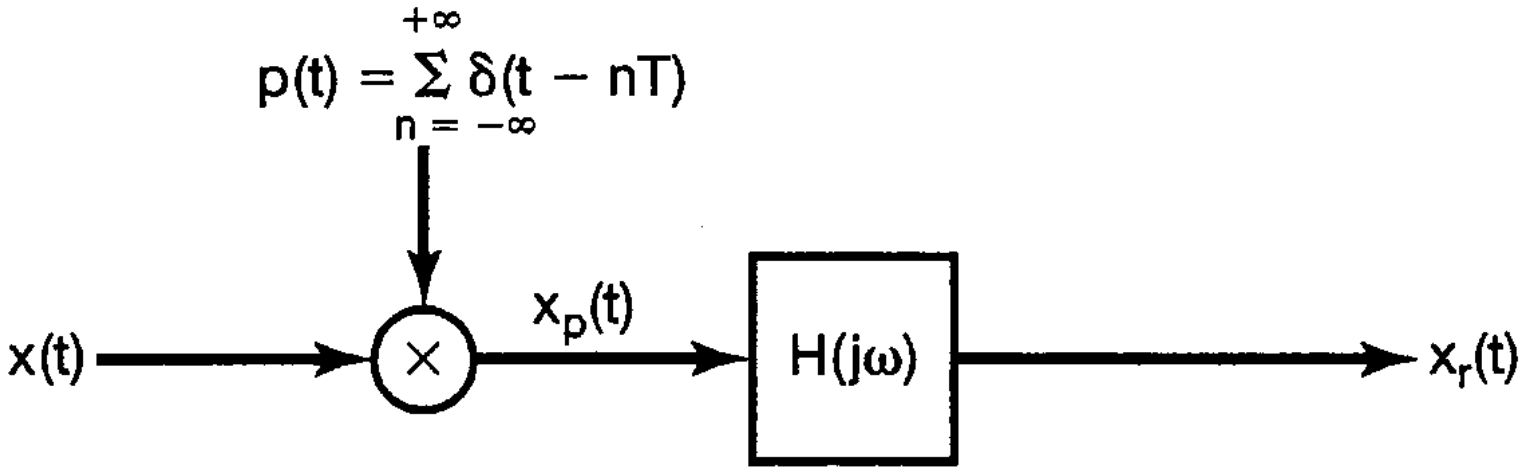
Ιδανική δειγματοληψία και ανακατασκευή στο πεδίο της συχνότητας

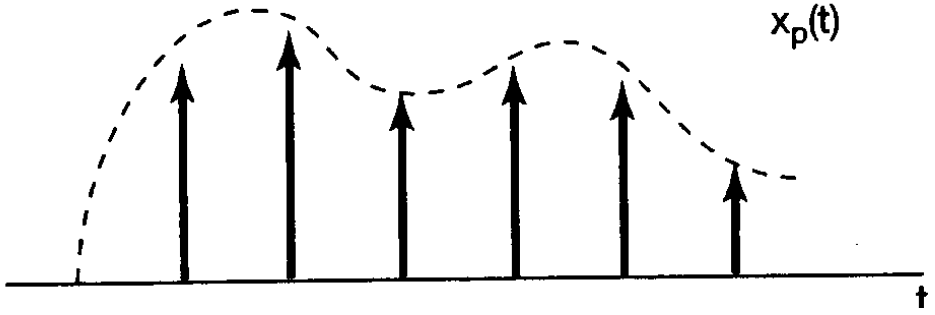


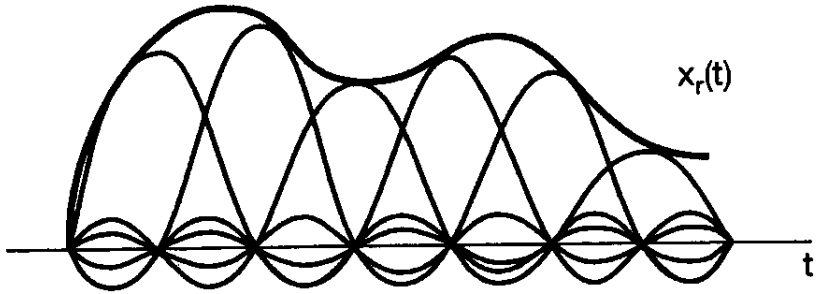
$$X_r(j\Omega) = H_r(j\Omega)X_s(j\Omega)$$

$$X_r(j\Omega) = H_r(j\Omega)X\left(e^{j\Omega T}\right) = \begin{cases} TX\left(e^{j\Omega T}\right), & |\Omega| < \pi/\mathrm{T} \\ 0, & \alpha\lambda\lambda ou \end{cases}$$

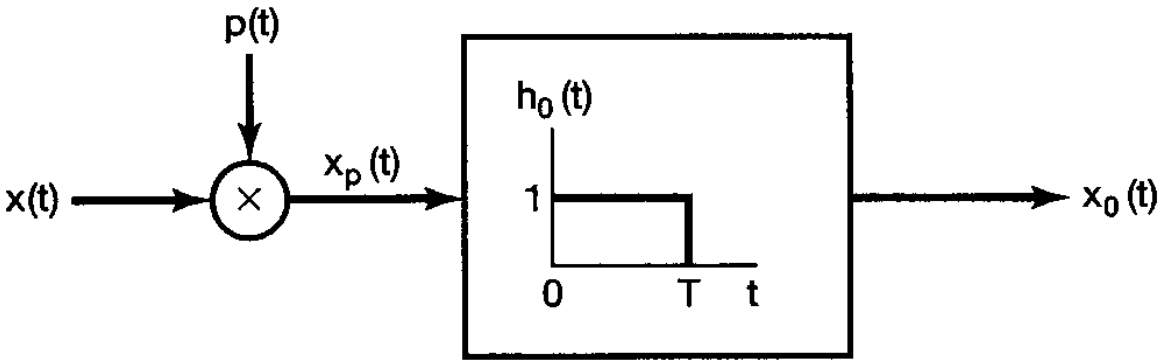
Ιδανική Παρεμβολή με Sinc $(\omega \rightarrow \Omega, \mathrm{p} \rightarrow s)$

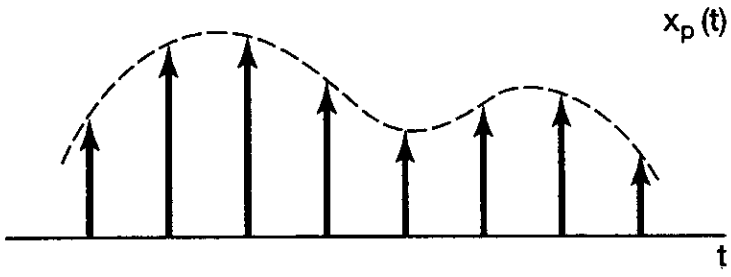


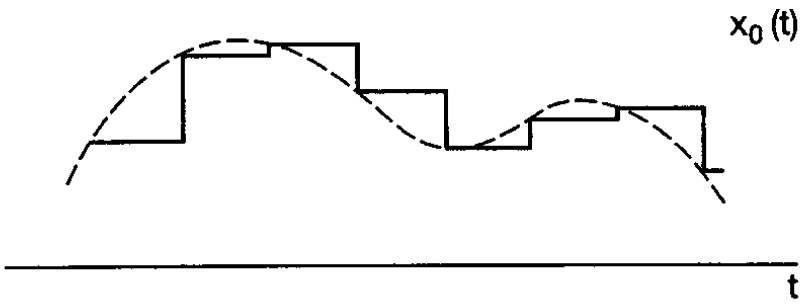


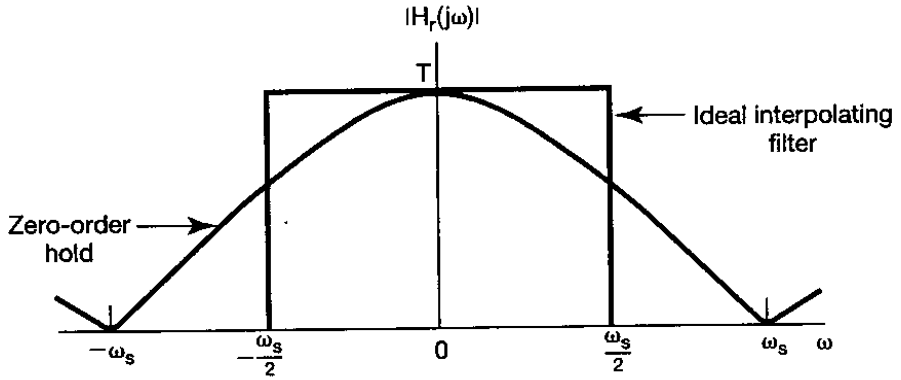


Zero-Order Hold Δειγματοληψία και Ανακατασκευή ($\omega \rightarrow \Omega, \mathrm{p} \rightarrow s$)

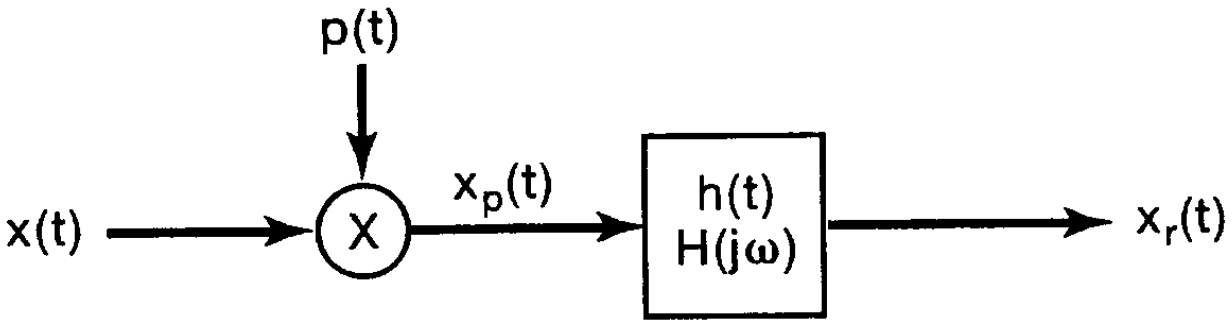


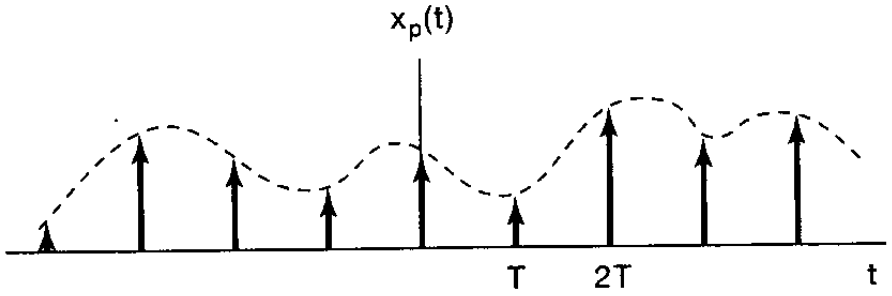


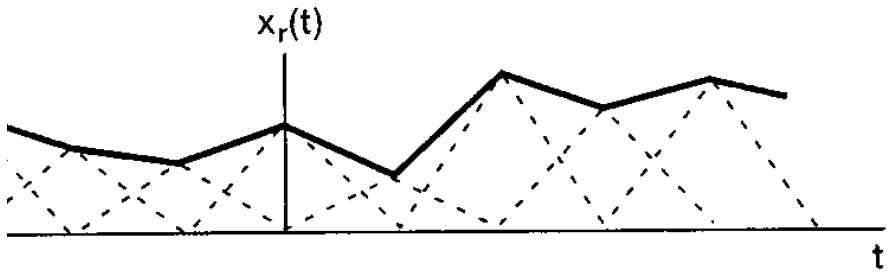


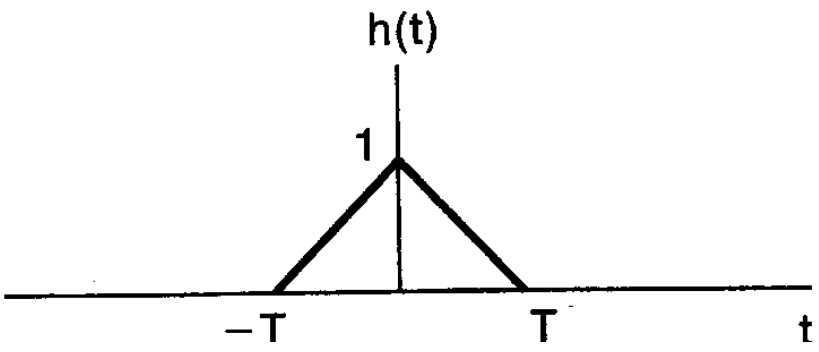


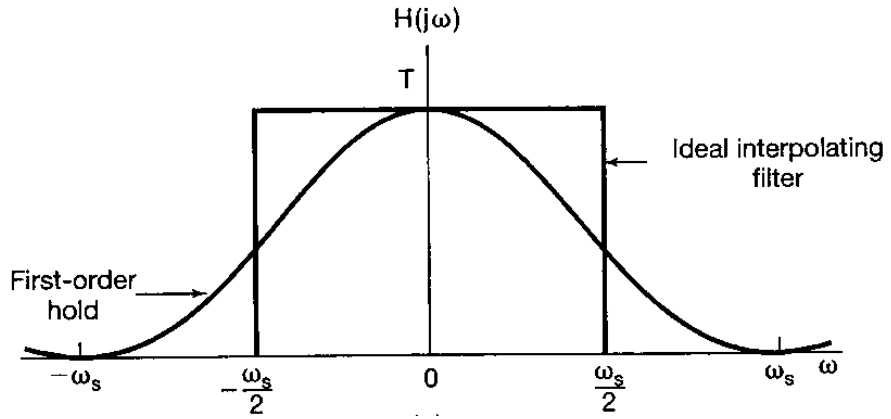
Γραμμική Παρεμβολή - 1st Order Hold ($\omega \rightarrow \Omega, \mathrm{p} \rightarrow s$)









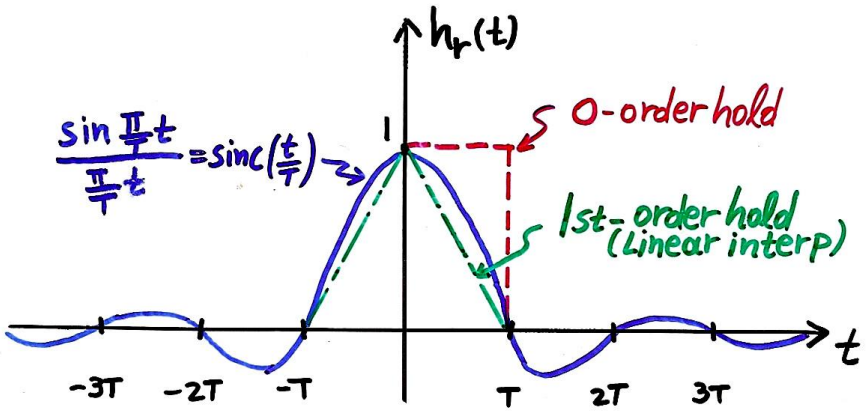


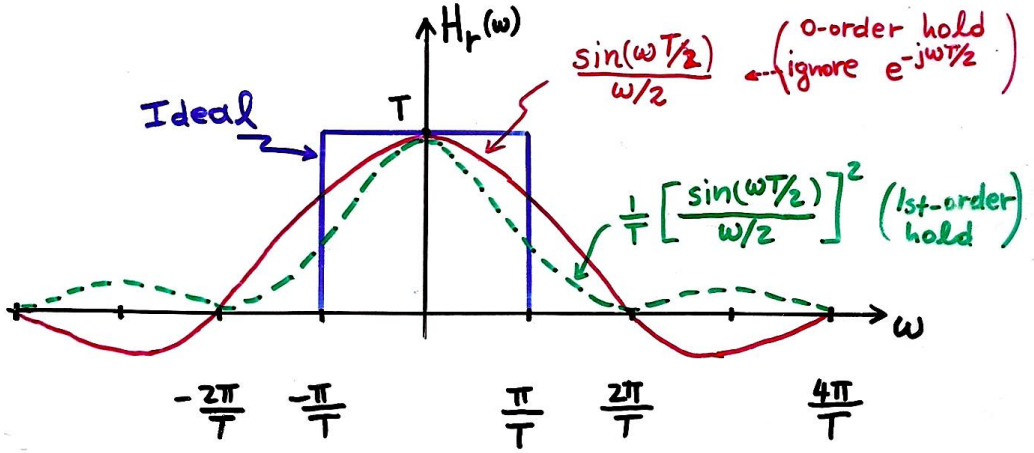
Ανακατασκευή σημάτων από δείγματα

$\omega \rightarrow \Omega$

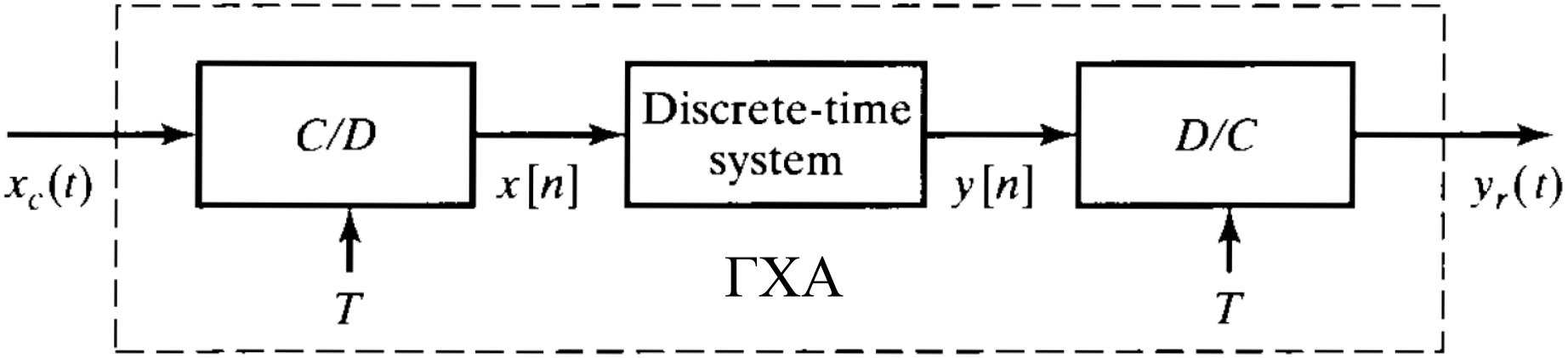
Reconstruction of sampled signals

$$x_r(t) = \sum_{n=-\infty}^{\infty} x(nT)\, h_r(t-nT)$$





Ψηφιακή Επεξεργασία Αναλογικών Σημάτων - I



C/D: Δειγματοληψία

$$x[n] = x_c(nT) \quad \overset{\mathcal{F}}{\leftrightarrow} \quad X(e^{j\omega}) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega}{T} - \frac{2\pi k}{T}\right)\right)$$

Ψηφιακό φίλτρο

$$y[n] = \sum_{k=1}^{N} a_k y[n-k] + \sum_{k=0}^{M} b_k x[n-k]$$

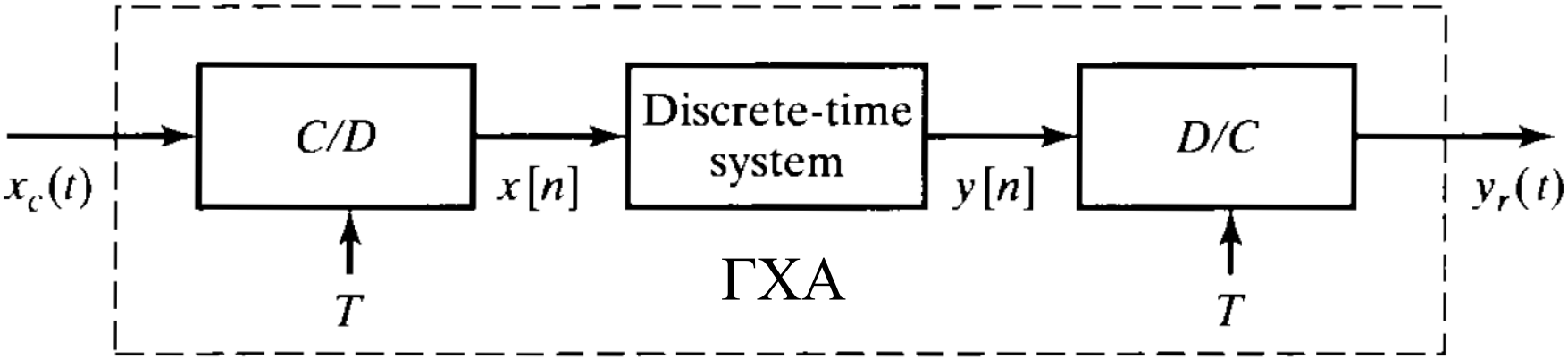
$$y_r(t) = \sum_{n=-\infty}^{\infty} y[n]\frac{\sin[\pi(t-nT)/T]}{\pi(t-nT)/T} \quad \overset{\mathcal{F}}{\leftrightarrow}$$

$$Y_r(j\Omega) = H_r(j\Omega)Y(e^{j\Omega T}) = \begin{cases} TY(e^{j\Omega T}), & |\Omega| < \pi/T, \\ 0, & \text{otherwise.} \end{cases}$$

D/C: Χρονική Παρεμβολή

D/C: Βαθυπερατό Φιλτράρισμα

Ψηφιακή Επεξεργασία Αναλογικών Σημάτων - ΙΙ



$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega})$$

$$Y_r(j\Omega) = H_r(j\Omega)H(e^{j\Omega T})X(e^{j\Omega T})$$

$$X(e^{j\Omega T}) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s))$$

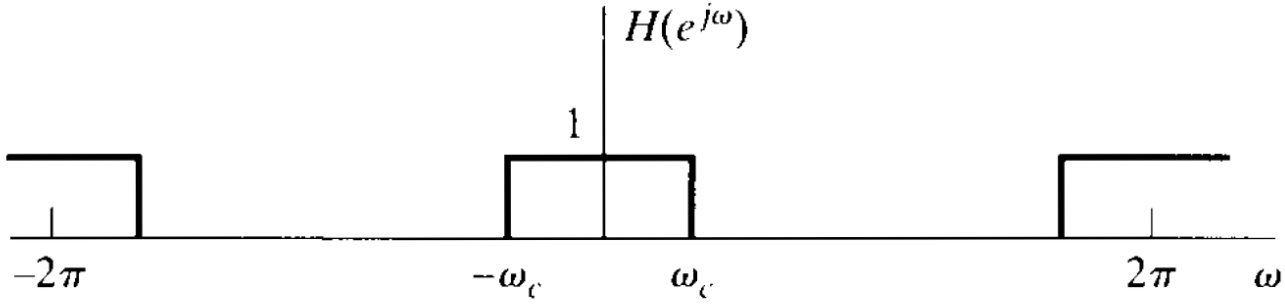
$$Y_r(j\Omega) = \begin{cases} H(e^{j\Omega T})X_c(j\Omega), & |\Omega| < \pi/T, \\ 0, & |\Omega| \geq \pi/T. \end{cases}$$

Αν ισχύουν οι συνθήκες του Θεωρήματος Δειγματοληψίας

$$Y_r(j\Omega) = H_{\text{eff}}(j\Omega)X_c(j\Omega)$$

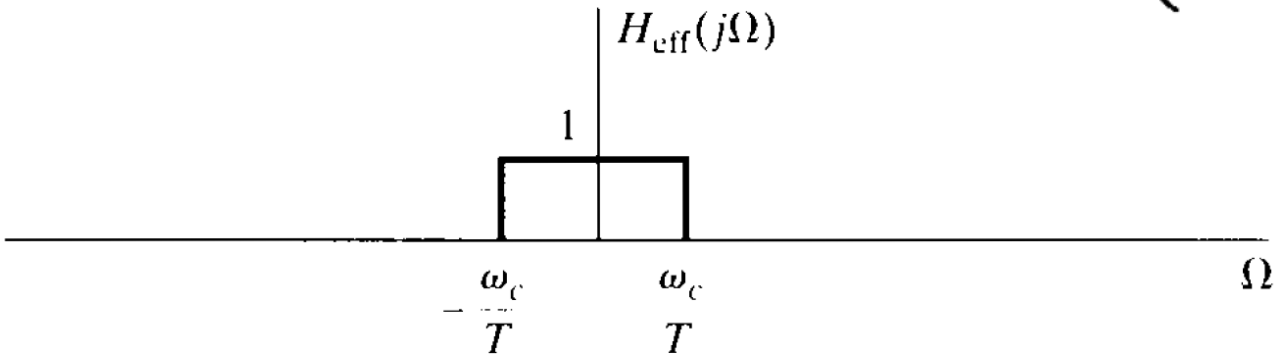
$$H_{\text{eff}}(j\Omega) = \begin{cases} H(e^{j\Omega T}), & |\Omega| < \pi/T, \\ 0, & |\Omega| \geq \pi/T. \end{cases}$$

Ιδανικό Αναλογικό Βαθυπερατό Φίλτρο από Ψηφιακό Βαθυπερατό



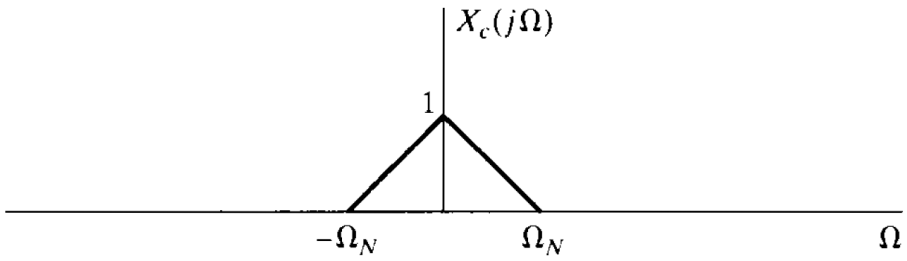
(a)

$$H(e^{j\omega}) = \begin{cases} 1, & |\omega| < \omega_c, \\ 0, & \omega_c < |\omega| \le \pi \end{cases}$$

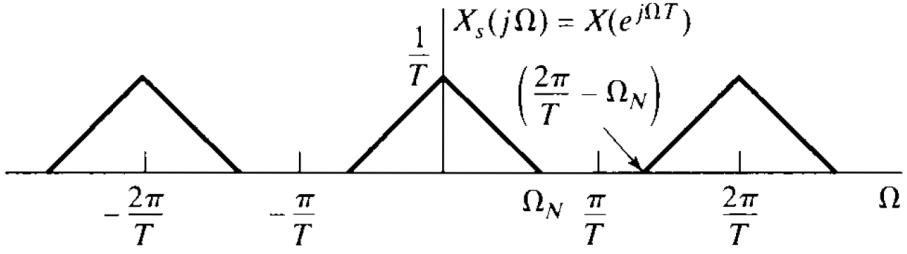


$$H_{\text{eff}}(j\Omega) = \begin{cases} 1, & |\Omega T| < \omega_c \text{ or } |\Omega| < \omega_c / T, \\ 0, & |\Omega T| > \omega_c \text{ or } |\Omega| > \omega_c / T. \end{cases}$$

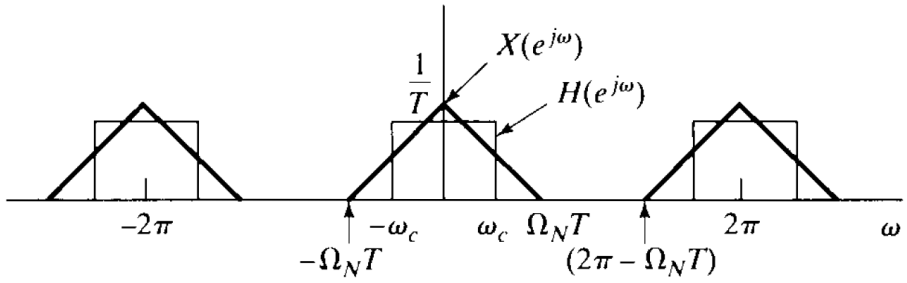
Παράδειγμα Ψηφιακής Επεξεργασίας Αναλογικού Σήματος



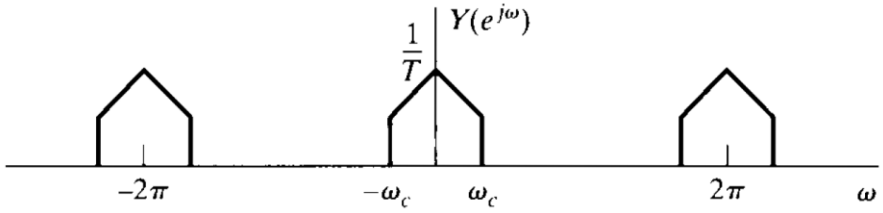
(a)



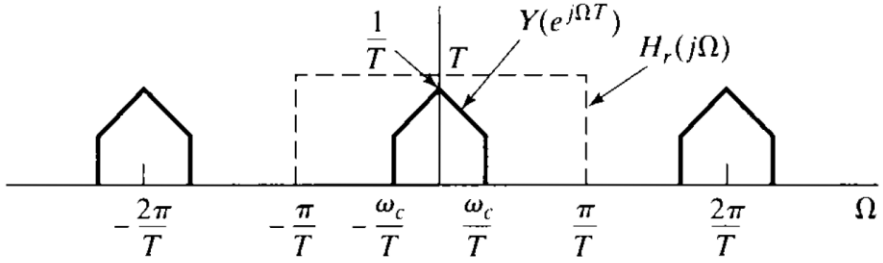
(b)



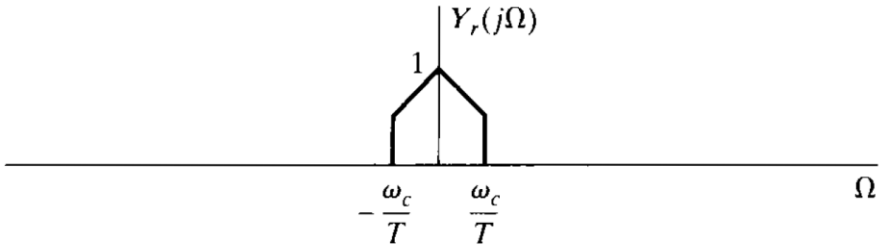
(c)



(d)



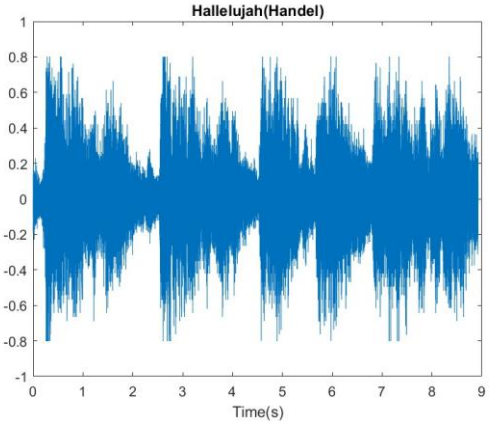
(e)

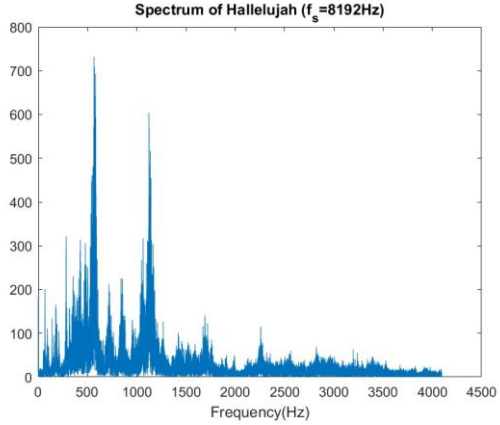


(f)

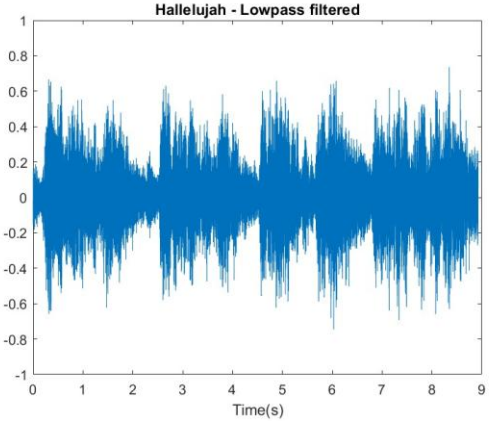
Ψηφιακή Επεξεργασία Μουσικού Σήματος

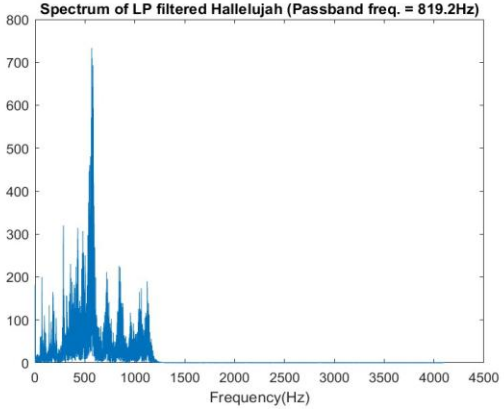




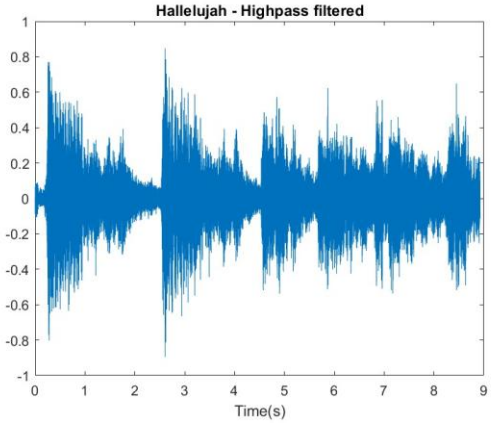


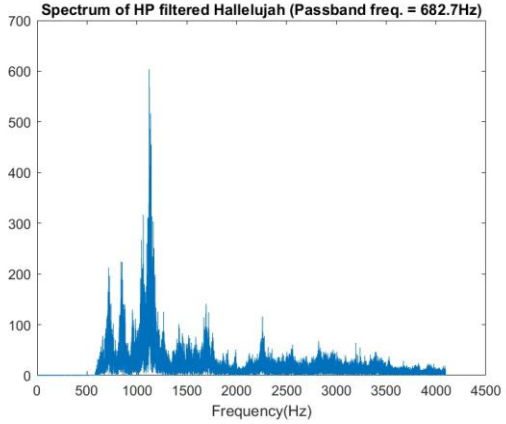




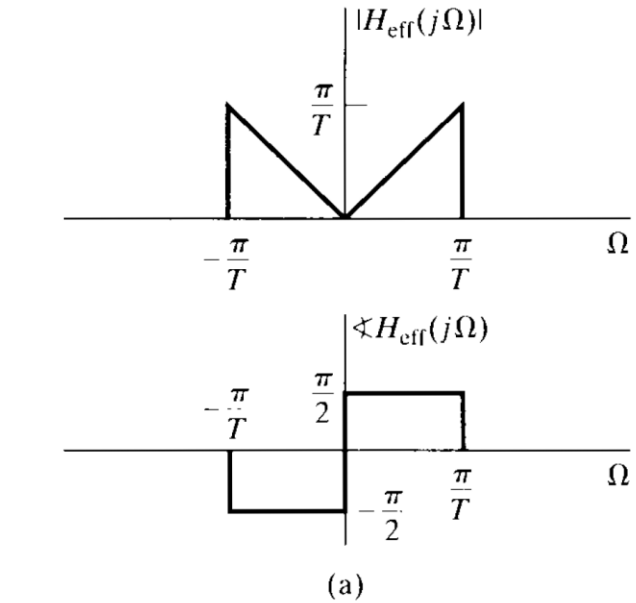








Ψηφιακή Υλοποίηση Ιδανικού Αναλογικού Bandlimited Διαφοριστή

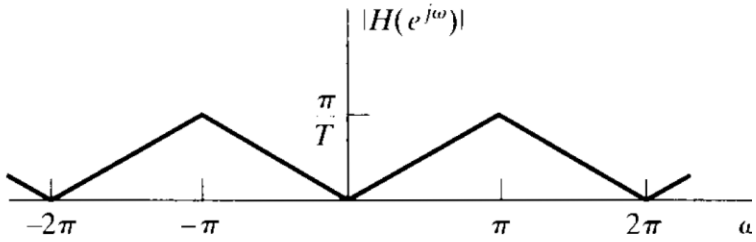


(a)

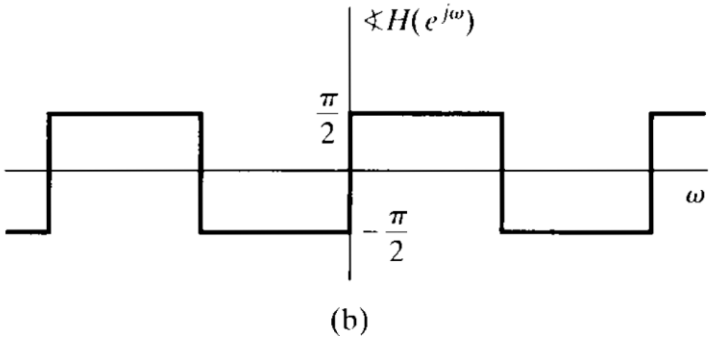
$$y_c(t) = \frac{d}{dt}[x_c(t)]$$

$$H_c(j\Omega) = j\Omega$$

$$H_{\text{eff}}(j\Omega) = \begin{cases} j\Omega, & |\Omega| < \pi/T, \\ 0, & |\Omega| \geq \pi/T, \end{cases}$$



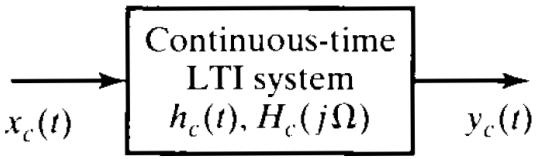
$$H(e^{j\omega}) = \frac{j\omega}{T}, \qquad |\omega| < \pi$$



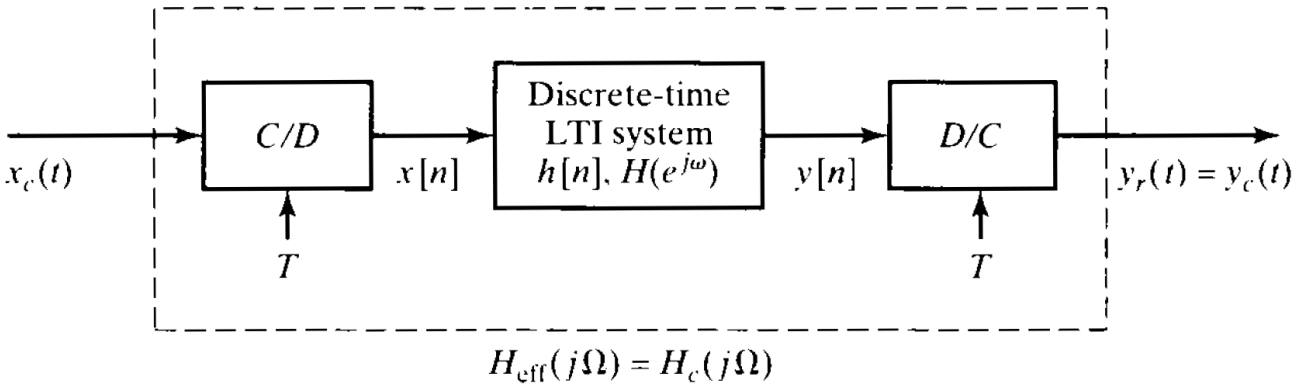
(b)

$$h[n] = \begin{cases} 0, & n = 0, \\ \dfrac{\cos \pi n}{nT}, & n \neq 0. \end{cases}$$

Αναλογικό ΓΧΑ Σύστημα και Ισοδύναμο Ψηφιακό για Bandlimited Σήματα Εισόδου



(a)



(b)

$$H(e^{j\omega}) = H_c(j\omega/T), \qquad |\omega| < \pi$$

$$H_c(j\Omega) = 0, \qquad |\Omega| \geq \pi/T$$

IMPULSE INVARIANCE: $h[n] = Th_c(nT)$

Βιβλιογραφία

[1] A. V. Oppenheim, R. W. Schafer with J.R. Buck, *Discrete-time Signal Processing,* 2nd ed., Prentice-Hall, 1999, Chapter 4.

[2] J. G. Proakis and D. G. Manolakis, *Ψηφιακη Ανάλυση Σήματος,* Prentice-Hall, 2007. Ελληνική έκδοση/μετάφραση, Εκδόσεις Ίων, 2010, Κεφ. 6.

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